Smart Water Quality Monitoring System

A Student Project Report

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Introduction: The Vital Need for Clean Water

Global Challenge

Access to safe drinking water is a fundamental human right, yet millions face contamination daily due to inadequate monitoring systems.

Traditional Limitations

Current lab-based water testing methods are slow, expensive, and impractical for continuous, real-time monitoring of water sources.

Intelligent Solution

This project presents a real-time, AI-powered monitoring system that detects contamination immediately and alerts stakeholders for rapid intervention.

Problem Statement: The Dangers of Unmonitored Water

1 Slow and Costly Testing

requires laboratory analysis, creating significant collection and results—often days or weeks.

2 Health Crisis

Contaminated water causes widespread waterborne diseases including cholera, dysentery, and typhoid, impacting millions of lives and livelihoods annually.

3 Remote Area Challenges

Lack of continuous monitoring infrastructure in remote or developing regions exacerbates contamination issues, leaving communities vulnerable without warning.

Project Objectives: What We Aim to Achieve

01

Low-Cost System Design

Develop an affordable, real-time water quality monitoring solution accessible to communities and organisations with limited resources.

02

Multi-Parameter Measurement

Simultaneously measure critical water quality indicators: pH levels, Total Dissolved Solids (TDS), temperature, and turbidity.

03

On-Device Intelligence

Integrate machine learning algorithms for intelligent impurity detection and classification directly on the device.

04

Cloud Dashboard

Enable remote data visualisation through an intuitive web-based dashboard accessible from anywhere.

05

Automated Alert System

Trigger immediate notifications when parameters exceed safe thresholds, enabling rapid response and intervention.

Methodology: Our Approach to Building the System

Hardware Design

Utilised the ESP32 microcontroller for efficient data collection, processing, and wireless communication capabilities.

Sensor Integration

Connected and calibrated multiple sensors for pH, TDS, temperature, and turbidity measurements with precise accuracy.

Machine Learning

Trained a neural network using TinyML on a custom dataset for real-time impurity classification and pattern recognition.

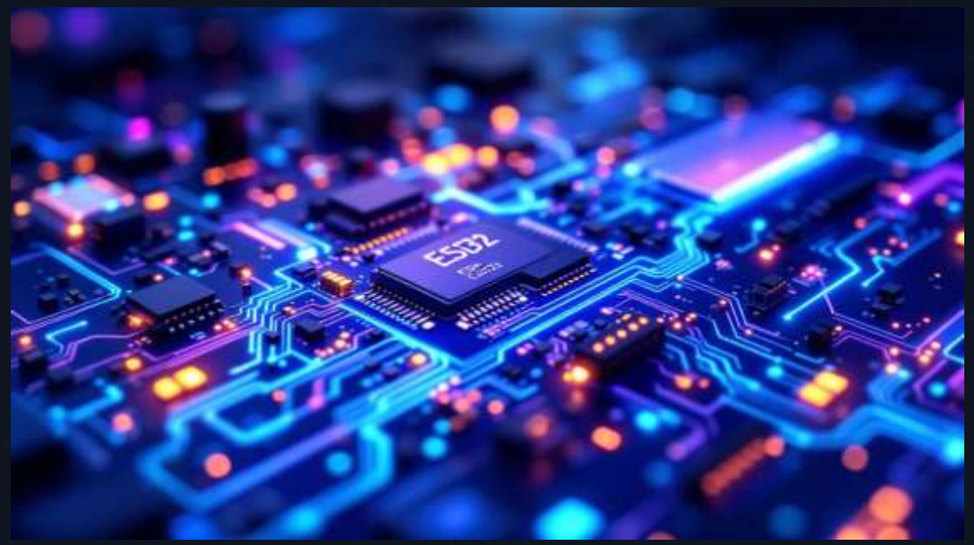
Cloud Connectivity

Established secure data transmission to cloud backend for storage, remote access, and dynamic visualisation through dashboards.

Alert Mechanism

Implemented automated notification system to alert users via mobile or web when critical water quality changes occur.

Technologies Used: The Tools of Our Trade



ESP32 Microcontroller

Chosen for superior processing power, dual Wi-Fi/Bluetooth capabilities, and cost-effectiveness for IoT applications.

Water Quality Sensors

Integrated pH sensor, TDS sensor, temperature sensor, and turbidity sensor for comprehensive parameter measurement.

TensorFlow Lite

Deployed TensorFlow Lite for Microcontrollers (TinyML) to run machine learning models directly on the ESP32 chip.

Firebase Platform

Utilised Firebase for secure cloud storage, real-time database, and dynamic web dashboard visualisation.

MQTT Protocol

Implemented MQTT for lightweight, efficient data transmission between device and cloud infrastructure.

How It Works: From Source to Insight

Continuous Sensing

Sensors deployed in water source continuously measure quality parameters in real-time.

Data Processing

ESP32 microcontroller collects sensor readings and processes them locally for immediate analysis.

ML Classification

On-device neural network categorises impurity events such as 'Normal', 'Rainwater Runoff', or 'Chemical Contamination'.

Cloud Transmission

Processed data transmitted securely to cloud platform for storage and dashboard display.

Visualisation

Dynamic web dashboard presents real-time status, historical trends, and interactive data visualisation.

Alert Trigger

Automated notifications sent immediately when parameters fall outside pre-defined safe thresholds.

Challenges Faced and Lessons Learned

Sensor Calibration

Challenge: Achieving consistent accuracy across multiple sensor types required extensive calibration and testing protocols.

Lesson: Learned importance of rigorous calibration procedures and environmental compensation techniques.

Power Management

Challenge: Optimising energy consumption for continuous 24/7 operation, especially in remote areas without reliable power sources.

Lesson: Mastered sleep modes, power-saving strategies, and battery management for IoT devices.

ML Model Deployment

Challenge: Fitting complex machine learning models onto resource-constrained microcontrollers with limited memory.

Lesson: Gained expertise in model quantisation, pruning, and optimisation techniques for embedded AI.

Data Security

Challenge: Protecting sensitive water quality data during transmission to cloud and preventing unauthorised access.

Lesson: Implemented encryption protocols, secure authentication, and data privacy best practices.

Applications and Future Scope: Beyond the Project

Current Applications

- **Household Safety**

  Real-time monitoring of home water supply for families and residential communities.

- **Industrial Control**

  Quality assurance in manufacturing, pharmaceuticals, and food processing facilities.

- **Environmental Monitoring**

  Tracking water quality in rivers, lakes, and coastal areas for ecological protection.

- **Smart Agriculture**

  Optimising irrigation systems and monitoring agricultural water quality for crop health.

Future Enhancements

- **Additional Sensors**

  Integration of heavy metal detectors, bacterial sensors, and advanced contaminant identification.

- **Autonomous Control**

  Development of automated pump and filtration systems based on ML predictions.

- **Predictive Maintenance**

  AI-driven forecasting of component failures and maintenance requirements for system reliability.

- **Network Expansion**

  Scalable deployment of distributed monitoring stations for city-wide or regional water quality networks.

Conclusion: A Step Towards Safer Water for All

Cost-Effective Solution

Our system provides an affordable, intelligent, and real-time water quality monitoring solution accessible to diverse communities.

Bridging Critical Gaps

Successfully addresses limitations of traditional water monitoring, significantly enhancing public health and safety through immediate detection.

IoT and AI Integration

This project demonstrates the transformative power of combining Internet of Things technology with artificial intelligence to solve real-world challenges.

> *"Empowering communities with intelligent water quality monitoring for a healthier, safer future."*